

Chapter 17 – Real-World Troubleshooting & Interview Scenarios

Learning Objectives

After this chapter, you should be able to:

- Break down real enterprise incidents systematically.
 - Identify root causes using logs and policies.
 - Apply troubleshooting frameworks under pressure.
 - Answer scenario-based interview questions confidently.
 - Connect multiple Workspace domains in one investigation.
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1. Core Troubleshooting Mindset

In enterprise environments, problems are rarely single-layer.

Every issue belongs to one or more categories:

1. Authentication

- Password
- MFA / 2SV
- Login challenges
- SAML SSO

2. Authorization

- Admin roles
- App permissions
- Sharing permissions

3. Context

- Context-Aware Access
- Device trust
- Location restrictions

4. Provisioning

- User creation
- SCIM sync

- Group membership

5. Application Layer

- Gmail
 - Drive
 - Calendar
 - Meet
 - Chat
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2. Standard Investigation Flow

```text id="flow1" Identify Issue

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Classify (Auth / Authz / Context / Provisioning / App)

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Check Logs

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Correlate Events

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Identify Root Cause

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Apply Fix

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Validate ```

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### 3. Scenario 1 – “Cannot Access Admin Console While Traveling”

#### Problem

An admin travels to Singapore and cannot access Admin Console.

Policy allows only India.

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#### Analysis

##### Step 1: Classification

✓ Context issue (CAA)

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##### Step 2: Likely Cause

- Location restriction = India only
  - Device might be non-compliant
  - Context-Aware Access blocking
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#### Correct Resolution

Do NOT disable security.

Options:

- Temporary CAA exception
  - Emergency admin access policy
  - Break-glass account (best practice)
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#### Key Interview Answer

“This is a Context-Aware Access restriction issue. I would verify the admin’s device and location context, and if necessary, use a controlled break-glass admin account or temporary policy exception rather than disabling security controls.”

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## 4. Scenario 2 – “Emails Not Receiving External Mail”

### Symptoms

- Internal email works
  - External email fails
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### Root Cause Path

1. Check MX records
  2. Check DNS propagation
  3. Check routing rules
  4. Check spam filtering
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### Most Likely Cause

- ✓ Incorrect or missing MX records
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## 5. Scenario 3 – “Drive File Suddenly Missing”

### Symptoms

- User cannot find file
  - No deletion notification
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### Investigation

Check:

- Drive audit logs
  - Shared Drive membership
  - Group membership changes
  - Ownership transfer logs
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## Likely Cause

- ✓ Removed from Shared Drive or group
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## 6. Scenario 4 – “Suspicious Login Detected”

### Signs

- New country login
  - New device
  - MFA prompt triggered
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### Investigation Flow

1. Login logs
  2. Device history
  3. OAuth grants
  4. Admin logs for password reset attempts
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### Possible Causes

- User traveling
  - Credential compromise
  - VPN usage
  - Password spraying attack
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## 7. Scenario 5 – “SPF / DKIM / DMARC Email Failure”

### Problem

Email rejected by recipient system

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### Investigation

Check:

- SPF record includes sending service?

- DKIM signing enabled?
  - DMARC policy strict?
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## Likely Causes

- Missing third-party sender in SPF
  - DKIM not configured
  - DMARC = reject policy too strict
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# 8. Scenario 6 – “OAuth App Access Suspicious”

## Problem

User installed unknown app

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## Investigation

Check OAuth logs:

- Scopes requested
  - User consent
  - App publisher
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## Risk

✓ Data exfiltration risk

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## Resolution

- Revoke app access
  - Block app in admin console
  - Educate users
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## 9. Scenario 7 – “Calendar Events Missing”

### Investigation

Check:

- Calendar audit logs
  - Delegation changes
  - External edits
  - Group-based calendar permissions
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### Likely Causes

- Deleted by delegate
  - Sharing removed
  - Sync issue
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## 10. Scenario 8 – “Users Locked Out After Policy Change”

### Root Cause

- Context-Aware Access misconfiguration
  - Chrome policy mismatch
  - Device compliance failure
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### Resolution

- Roll back policy
  - Use emergency admin account
  - Validate device trust rules
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## 11. Scenario 9 – “Drive External Sharing Incident”

### Problem

Sensitive file shared externally

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## Investigation

Use:

- Drive logs
- Investigation Tool
- Audit logs
- Trusted domain settings

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## Root Cause

- User manually shared file externally
- No DLP rule blocking action

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# 12. Scenario 10 – “Migration Failed Midway”

## Investigation

Check:

- Tool used (GWMME / DMS / CloudM)
- Network interruption
- Authentication failure
- API quota limits

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## Resolution

- Resume incremental migration
- Validate batch completion
- Re-run failed mailbox sync

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# 13. Advanced Interview Thinking Pattern

Always structure answers like this:

## 1. Identify domain

(Gmail / Drive / Security / Identity)

## 2. Identify category

(Auth / Authz / Context / Provisioning / App)

## 3. Identify logs

(Admin / Drive / Gmail / OAuth)

## 4. Identify policy impact

(CAA / DLP / Sharing / Admin rules)

## 5. Propose fix

(Least disruptive, secure approach)

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# 14. Common Interview Mistakes

- ✗ Jumping to solutions without logs
  - ✗ Disabling security policies immediately
  - ✗ Ignoring Context-Aware Access
  - ✗ Not considering OAuth risks
  - ✗ Forgetting group-based access impact
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# 15. Key Takeaways

- Every issue fits into a structured category.
  - Logs are your primary source of truth.
  - Policies (CAA, DLP, sharing rules) are often the root cause.
  - Fixes must be secure, not just quick.
  - Enterprise thinking = minimal disruption + maximum control.
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# Chapter Summary

In this chapter you learned:

- ✓ Real enterprise troubleshooting scenarios
- ✓ Structured investigation methodology
- ✓ Authentication vs authorization vs context breakdown
- ✓ Log-based root cause analysis

- ✓ Interview-ready answer frameworks
  - ✓ End-to-end incident thinking
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## What Comes Next (Final Phase)

Now we move into the final stage of mastery:

- 🔥 **Mock Interview Simulation (real pressure questions)**
- 🔥 **100+ scenario-based questions (Google Workspace Engineer level)**
- 🔥 **Architecture design challenges**
- 🔥 **API + automation scenarios**
- 🔥 **Real incident “war stories” (what happens in production)**

This is where you transition from learner → interview-ready engineer.